

IMPERIAL, NE, USA

WMO: 725626

Lat: 40.510N

Lon: 101.620W

Elev: 3268

StdP: 13.04

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24091

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.2	4.8	-7.5	4.1	1.8	-2.5	5.4	7.6	33.3	31.9	29.4	30.7	8.3	250	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.9	99.1	67.5	96.2	67.4	92.7	67.0	72.4	87.6	71.0	86.6	69.7	85.4	12.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	117.7	77.1	66.5	110.2	76.1	65.2	105.3	75.7	38.2	87.3	36.8	86.2	35.7	85.4	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.6	25.9	23.0	DB	-8.6	104.4	6.0	3.1	-13.0	106.6	-16.5	108.4	-19.9	110.2	-24.3	112.4	(4)
(5)			WB	-9.3	75.3	5.9	1.3	-13.5	76.2	-16.9	77.0	-20.2	77.7	-24.5	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	29.5	30.4	41.4	49.5	59.4	71.4	77.4	74.4	66.0	51.6	39.1	29.2
	DBStd	19.43	10.87	11.37	10.27	9.11	8.98	6.92	5.33	5.63	8.14	9.30	9.92	10.58
	HDD50	2704	635	550	297	117	24	0	0	0	4	90	341	646
	HDD65	5904	1100	968	732	467	217	22	2	5	84	420	776	1111
	CDD50	3338	1	2	31	101	315	641	849	757	485	141	15	0
	CDD65	1063	0	0	0	2	44	213	386	297	115	6	0	0
	CDH74	13349	0	1	48	169	778	2650	4586	3263	1592	252	10	0
	CDH80	6634	0	0	7	39	295	1275	2554	1659	733	71	1	0
(14) Wind	WSAvg	10.5	10.2	10.7	11.5	12.7	11.7	10.7	9.3	8.9	9.6	10.1	10.0	10.0
(15) Precipitation	PrecAvg	19.5	0.5	0.5	1.2	1.8	3.0	3.3	2.9	2.4	1.5	1.3	0.7	0.4
	PrecMax	29.7	1.7	2.2	6.1	6.7	7.6	8.7	7.0	8.6	6.3	4.2	2.0	1.3
	PrecMin	10.4	0.0	0.0	0.1	0.1	0.1	0.6	0.5	0.1	0.0	0.0	0.0	0.0
	PrecStd	4.0	0.4	0.5	1.1	1.3	1.7	1.9	1.5	1.7	1.3	1.1	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.3	69.9	80.2	85.2	93.3	100.4	104.1	100.4	97.8	87.3	75.1	63.2
		MCWB	44.9	47.1	52.5	56.1	61.5	66.4	67.5	67.1	63.2	58.8	52.1	44.6
	2%	DB	57.7	61.0	72.9	79.1	87.9	96.8	99.9	97.0	92.8	81.2	68.6	56.0
		MCWB	42.5	43.7	50.1	55.1	61.9	66.5	68.1	67.8	64.2	56.6	49.0	41.6
	5%	DB	51.7	54.4	66.9	73.4	82.3	91.8	96.9	93.2	88.8	75.2	62.8	50.5
		MCWB	39.5	40.9	47.8	53.3	60.3	66.5	68.5	67.6	63.2	54.8	46.9	38.9
	10%	DB	45.8	48.4	60.9	67.9	77.7	87.6	93.1	89.7	83.6	69.5	56.3	45.1
		MCWB	36.2	38.2	45.5	51.3	58.6	65.8	68.8	67.6	62.2	52.9	43.6	35.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	48.3	56.1	59.7	67.3	73.2	74.7	73.4	69.4	61.9	54.0	46.1
		MCDB	62.3	67.5	71.3	77.0	83.7	87.6	89.0	88.5	85.7	77.3	68.9	60.6
	2%	WB	42.6	44.3	52.3	57.1	64.4	70.8	72.9	71.6	67.5	59.2	50.5	42.3
		MCDB	56.6	59.2	68.5	72.8	79.7	85.7	88.0	87.1	83.4	73.9	65.5	54.8
	5%	WB	39.8	41.4	49.2	54.7	62.4	69.0	71.4	70.3	65.9	56.7	47.5	39.2
		MCDB	50.5	53.6	64.9	69.7	77.4	84.4	87.2	85.7	81.0	71.2	61.6	49.9
	10%	WB	36.8	38.4	46.3	52.5	60.3	67.6	70.3	69.1	64.2	54.2	44.3	36.1
		MCDB	45.3	48.1	60.0	66.4	74.2	83.7	86.2	84.1	78.4	66.9	56.0	44.9
(35) Mean Daily Temperature Range	5% DB	MDBR	24.5	23.9	27.9	26.5	25.5	27.2	27.9	27.0	28.8	27.5	26.5	23.9
		MCDBR	33.3	34.7	39.2	37.7	33.9	33.3	33.9	33.0	35.7	38.2	36.5	31.9
	5% WB	MCWBR	20.1	20.1	20.2	17.7	13.2	11.1	10.2	10.3	12.1	17.6	20.0	19.8
		MCWBR	31.5	33.1	35.8	32.7	29.3	29.1	28.4	28.0	29.4	32.0	34.6	31.0
(40) Clear-Sky Solar Irradiance	taub		0.256	0.264	0.290	0.317	0.344	0.357	0.379	0.369	0.335	0.296	0.266	0.256
		taud	2.554	2.532	2.477	2.424	2.420	2.416	2.400	2.413	2.473	2.547	2.552	2.536
	Ebn at Noon		296	307	308	304	296	290	283	283	287	291	288	286
		Edh at Noon	22	26	31	35	36	36	37	35	31	26	22	21
(44) All-Sky Solar Radiation	RadAvg		758	1032	1409	1735	2009	2308	2276	1985	1614	1153	826	657
	RadStd		54	77	114	95	137	115	86	92	72	106	52	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page